



DETAILED REPORT

Brick Destroyer – JavaFX Game Application

1. Project Title

Brick Destroyer – JavaFX Game Application

2. Project Description

Brick Destroyer is a JavaFX-based desktop game application. In this game, the player controls a paddle using the mouse to bounce a ball and destroy bricks on the screen.

The game includes multiple difficulty levels, a login system, score tracking, and file saving features. The main objective is to destroy all bricks without losing all balls.

3. Project Features

3.1 Game Features

- Paddle controlled by mouse movement
 - Ball movement with bounce physics
 - Brick destruction system
 - Score system (10 points per brick)
 - Multi-ball system (new ball appears after score milestones)
 - Win and lose conditions
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3.2 Authentication System

- User login and registration system
 - User data stored in `users.txt` file
 - If user does not exist, system automatically creates a new account
 - Password verification for existing users
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3.3 File Handling

- High score saved in `highscore.txt`
 - Previous high score is loaded when the application starts
 - Score history can be reset
 - Uses Java `FileWriter` and `Scanner` classes
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3.4 Difficulty Levels

- Recruit (Easy speed)
- Veteran (Medium speed)
- Elite (Hard speed)

Each difficulty level changes the speed of the ball.

3.5 Graphical User Interface (GUI)

- Built using JavaFX
 - Dark gradient background design
 - Styled buttons and input fields
 - Drop shadow effects for UI elements
 - Fullscreen support (F11 key)
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4. System Design (How It Works)

Step 1: Login Screen

The user enters a username and password. The system checks the `users.txt` file.

- If the user exists → login successful
 - If the user does not exist → a new account is automatically created
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Step 2: Level Selection

The user selects a difficulty level:

- Recruit (Slow speed)

- Veteran (Medium speed)
 - Elite (Fast speed)
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Step 3: Game Start

- Paddle is created at the bottom of the screen
 - One ball is created
 - Bricks are generated in a grid layout
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Step 4: Game Loop

The game runs using `AnimationTimer`:

- Ball moves continuously
 - Collision detection is applied: wall, paddle, and bricks
 - Bricks disappear when hit
 - Score increases by 10 for each brick destroyed
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Step 5: Game End

The game ends when:

- All bricks are destroyed → Victory
- All balls are lost → Game Over

If a new high score is achieved, it is saved in a file.

5. Technologies Used

- Java
- JavaFX
- Object-Oriented Programming (OOP)
- File Handling (Java IO)
- `AnimationTimer` (Game Loop)